Rolling Bearing and Vibration Severity Guide

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Vibration severity limits for pumps up to 15 kW

Zone	Velocity RMS (mm/s)	Condition
A	below 2.8	New machine
B	2.8 to 4.5	Acceptable for long-term operation
C	4.5 to 7.1	Unsatisfactory, plan maintenance
D	above 7.1	Damage likely, stop machine

Common bearing failure modes

Inadequate lubrication, contamination, misalignment, overload and fatigue spalling. Typical symptoms include increased high-frequency vibration and rising bearing temperature.

Lubrication

Grease intervals depend on speed, temperature and bearing size. For deep groove ball bearings at 1500 rpm and 70 deg C housing temperature, a typical interval is 3000 to 5000 operating hours.